

Hardik Pandey

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EDUCATION

Vellore Institute of Technology

Bhopal, India

Bachelor of Technology in Computer Science and Engineering; CGPA: 9.16/10.0

Aug. 2023 – May 2027

TECHNICAL SKILLS

Programming Languages: Python, Java, JavaScript, Go, SQL, Ruby, C/C++ (familiar)

Data Structures & Algorithms: Object-oriented design, algorithm design, complexity analysis, problem solving, optimization

Distributed Systems: Microservices architecture, scalable system design, fault-tolerant systems, multi-tiered systems

Backend Development: Node.js, FastAPI, Flask, RESTful APIs, asynchronous processing, high-throughput services

Databases: PostgreSQL, MySQL, MongoDB, Redis, query optimization, relational database design, distributed storage

DevOps & Cloud: AWS, Docker, Git, GitHub Actions, CI/CD pipelines, Linux, containerization, monitoring

Software Development: Agile methodologies, test-driven development, automated testing (Pytest, Jest), code review

PROJECTS

Ascraa – AI-Powered Verification Platform | *Python, FastAPI, PostgreSQL, Docker, React* Jan. 2025 – Present

- Designed and built scalable distributed system processing 10,000+ daily API requests using microservices architecture with FastAPI; implemented fault-tolerant design patterns and load balancing for high availability
- Developed comprehensive automated testing framework achieving 90%+ code coverage using Pytest; established CI/CD pipeline with GitHub Actions enabling rapid iteration cycles and continuous deployment
- Optimized PostgreSQL database queries from 2s to sub-200ms through indexing strategies, query planning, and connection pooling; designed normalized schema supporting concurrent read/write operations at scale
- Built real-time monitoring and alerting system tracking system performance, error rates, and user behavior patterns; implemented asynchronous task processing for improved system responsiveness and throughput

SentinelX – Social Media Analytics Platform | *Python, FastAPI, PostgreSQL, Docker* Oct. 2024 – Present

- Architected high-performance distributed API service handling 100,000+ daily requests using asynchronous processing and horizontal scaling; designed system to be fault-tolerant with graceful degradation and error recovery
- Implemented data processing pipeline with complex algorithm design for pattern recognition and anomaly detection; optimized computational efficiency reducing processing time by 75% through algorithmic improvements
- Developed automated testing suite covering unit, integration, and load testing scenarios; collaborated in agile environment with weekly sprints delivering incremental feature releases and rapid problem-solving
- Reduced query execution time by 80% through database optimization including proper indexing, query rewriting, and denormalization strategies; built scalable data storage layer supporting millions of records

AyushSetu – Healthcare Data Platform | *Python, Flask, PostgreSQL, REST APIs* Sep. 2024 – Present

- Built production-ready RESTful API service managing 50,000+ records with focus on data integrity, ACID compliance, and complex relational database operations; designed normalized schema with proper foreign key constraints
- Engineered multi-tiered system architecture separating concerns across presentation, business logic, and data access layers; implemented object-oriented design principles with clear abstractions and interfaces
- Developed automated testing framework validating business logic, data integrity, and API contracts; utilized test-driven development approach improving code quality and reducing defects by 40%
- Optimized database performance reducing query execution from 3s to 300ms through index optimization, query analysis, and database tuning; implemented caching layer with Redis for frequently accessed data

ACHIEVEMENTS & CONTINUOUS LEARNING

Algorithmic Problem Solving – Solved 200+ data structures and algorithms problems focusing on complexity analysis, optimization techniques, and efficient algorithm design; strong foundation in computer science fundamentals

Applied Machine Learning in Python – University of Michigan (Coursera) – Completed comprehensive course covering supervised/unsupervised learning, model evaluation, and practical applications of ML algorithms